MCAST

ASSESSMENT AND INTERNAL VERIFICATION FRONT SHEET (Individual Criteria)

(Note : This version is to be used for an assignment brief issued to students via Classter)

Course Title	**Bachelor of Science (Honours) Creative Computing/ BSc (Hons) Digital Games Development**		**Lecturer Name & Surname**	**Roderick Vella Galea**	
Unit Number & Title	ITMSD-606-2303: XR Development				
Assignment Number, Title / Type	Assignment 1 – Home				
Date Set		**Deadline Date**	Inside Document		
Student Name		**ID Number**		**Class / Group**	

Assessment Criteria	**Maximum Mark**
R&U1: Examine the process needed for a working XR environment.	5
R&U3: Identify design choices when turning a concept into working XR code.	5
R&U4: Identify input data that must be captured for a chosen interaction.	5
R&U5: Identify the key elements needed for a simple XR experience.	5
R&U6: Describe common methods that keep users engaged in XR.	5
R&U7: Recognise principles of feedback that motivate players in XR games	5
A&A2: Implement spatial input so a user can grab, press or move a virtual item.	7
A&A3: Investigate the user experience while the application is running.	7
A&A4: Implement the core logic of the designed XR experience	7
A&A5: Experiment with variations of game elements to create distinctive gameplay experiences.	7
E&C2: Assemble a polished XR prototype that runs smoothly and meets defined quality targets on the chosen XR hardware.	10
E&C3: Develop a coherent game-code structure that supports the incorporated mechanics and engagement goals.	10
Total Marks	78

Notes to Students:

- This assignment brief has been approved and released by the Internal Verifier through Classter.
- Assessment marks and feedback by the lecturer will be available online via Classter (Http://mcast.classter.com) following release by the Internal Verifier
- Students submitting their assignment on Moodle/Unicheck will be requested to confirm online the following statements:

Student's declaration prior to handing-in of assignment

❖ I certify that the work submitted for this assignment is my own and that I have read and understood the respective Plagiarism Policy

Student's declaration on assessment special arrangements

❖ I certify that adequate support was given to me during the assignment through the Institute and/or the Inclusive Education Unit.

❖ I declare that I refused the special support offered by the Institute.

ITMSD-606-2303: XR Development

Assignment - Home

MCAST - BSc (Hons) Creative Computing / BSc (Hons) Digital Games Development

General Guidelines:

• This assignment carries 78% of the global mark.

• Read the scenario carefully and follow the instructions given

• Any reference from class notes, books or internet should be included in the appendix

• Plagiarism is strictly prohibited and will be penalised in accordance with the college's disciplinary procedures.

• Your lecturer will provide you with details to upload your work.

• Late assignments are not accepted

• Read carefully the rubrics

Teamwork Section 1 Deadline -> 16th January 2026

Individual Section 2 Deadline -> 30th January 2026

Section 1 – Teamwork

Task 1

In this team project, you are required to design and develop a serious game using Unity XR that simulates a real-world process in medical or engineering contexts (e.g., performing a medical procedure, assembling machinery, maintaining equipment, etc) OR an entertainment game. This project is team-based but with clear individual responsibilities. Each student must demonstrate technical and reflective contributions to the final product. A proposal must be submitted and accepted by the lecturer.

Requirements

- As a team, submit a proposal including:
    - Game title and short description of the simulation/entertainment game concept.
    - Clear list of XR functionalities assigned to each student (minimum two per student). Functionality must have code behind it.
    - Brief description of how feedback mechanisms will guide or instruct the player (visual, audio, textual).
- Each student must:
    - Help in the design of the environment.
    - Develop at least two XR functionalities.
    - Contribute actively to team integration, testing, and refinement.
- The system must include a feedback mechanism that explains what the user needs to do (e.g., prompts, highlights, tooltips, or voice instructions).
- Each student must compile an individual progress report (journal), describing weekly contributions with screenshots, code snippets, and any challenges encountered. Compile into a single shared document. Make sure you separate by student name.
- The complete game must be compiled and demonstrated on the Meta Quest headset in front of the class.

Teamwork Regulations

- **If a student is not cooperating with their team members, then the lecturer has the right to remove that student from the team and assign them to do the project on their own, with a significant mark deduction. In case a student is going to miss a lecture during the assigned session, they must inform their lecturer and their team by 9 AM.**

R&U7: Recognise principles of feedback that motivate players in XR games. E&C2: Assemble a polished XR prototype that runs smoothly and meets defined quality targets on the chosen XR hardware. **Rubric 1: Game Quality (Team Assessment)**				
	PROFICIENT [11-15 Marks]	NEARING PROFICIENT [5-10 Marks]	NOVICE [1-4 Marks]	INADEQUATE [0 MARKS]
[15 Marks]	Game demonstrates a professional level of design and interactivity. Gameplay is smooth, immersive, and technically stable. Feedback mechanisms effectively guide the user through tasks. Visuals and environment are coherent and well-integrated. Demonstration shows strong teamwork and understanding of XR user experience.	Game shows clear educational purpose and functional design. Most interactions work as intended, with minor technical or usability issues. Feedback mechanisms provide reasonable guidance. Visual design is adequate and consistent. Demonstration is organised and shows collaborative effort.	Basic functionality achieved with limited realism or interactivity. Educational purpose is present but weakly expressed. Feedback mechanisms are minimal or unclear. Visual and technical polish limited. Demonstration functional but lacks depth.	Missing or irrelevant work

A&A4: Implement the core logic of the designed XR experience **Rubric 2: Individual Functionalities & Blender Object**		
Functionality ID	**Description**	**Max Mark**
F1	First XR functionality implemented correctly.	3.5
F2	Second XR functionality implemented correctly.	3.5

R&U3: Identify design choices when turning a concept into working XR code. **Rubric 3: Individual Progress Report**				
	PROFICIENT [5 Marks]	NEARING PROFICIENT [3-4 Marks]	NOVICE [1-2 Marks]	INADEQUATE [0 MARKS]
[5 Marks]	Report is detailed, well-structured by week, and includes screenshots, code excerpts, and clear reflection on each contribution. Demonstrates deep understanding of the teamwork process, problem-solving, and learning outcomes.	Report summarises weekly contributions clearly with some technical evidence (screenshots/code). Reflection is general but shows awareness of challenges and improvements.	Report includes basic weekly notes or screenshots with limited explanation. Minimal reflection on work or learning.	Missing or irrelevant work

Section 2 – Individual

Task 1

In this task, you are to review and analyse recent academic research (last 5 years) on Extended Reality (XR) development (including VR, AR, or MR environments). Your goal is to understand the process needed to create a working XR environment, from design to deployment.

You are required to:

- Search for **at least five (5)** peer-reviewed academic papers (in the last five years) related to XR development, workflow, or production pipelines.
- Extract key insights on tools, engines, SDKs, hardware, interaction methods, optimisation, and testing.
- Identify common stages of XR production (e.g., design, prototyping, asset creation, integration, testing, deployment).
- Build a clear conclusion describing what current research shows about modern XR workflows and what challenges developers face.

All findings must be compiled into a video PowerPoint presentation (8-10 minutes):

- Present your slides while your **camera and screen are both on**.
- When quoting or referencing, **show the highlighted text from the PDF page** to demonstrate authenticity.
- Discuss your analysis clearly
- Include a reference list in **Harvard** style.

R&U1: Examine the process needed for a working XR environment.				
	PROFICIENT [4-5 Marks]	NEARING PROFICIENT [2-3 Marks]	NOVICE [1 Mark]	INADEQUATE [0 MARKS]
[5 Marks]	Clearly identifies and compares multiple XR development stages and tools using well-selected academic papers. Demonstrates a strong understanding of the XR workflow, from setup to deployment, with evidence of synthesis and critical insight. Presentation is well-structured and professional, showing quoted pages on screen.	Describes several XR stages and tools with some supporting evidence. Demonstrates partial understanding of workflow but lacks consistent synthesis or evaluation. Presentation is generally clear with some proof of sources.	Mentions XR concepts without coherent structure or limited references. Shows minimal understanding of workflow or outdated sources. Presentation lacks clarity or visual proof of sources.	Missing or irrelevant work

Task 2

In this task, you are to identify and apply the key elements required to create an effective XR experience and keeping the users engaged. Choose one type of XR technology (VR, AR, or MR) and imagine you are the lead designer/developer creating a new product that offers a high-quality immersive experience.

You are required to:

- Research usability and design guidelines for XR experiences using reputable books, academic papers, and professional sources
- Select or invent a realistic scenario (e.g., VR safety training, AR tourism app, MR collaborative workspace).
- Apply the usability principles and design guidelines you found to your chosen scenario. Explaining how they influence your design choices and how it improves the user engagement.
- Produce sketches, storyboards, interface mock-ups, or UML diagrams to illustrate interaction flow, environment layout, or system logic.
- Compile everything into a video PowerPoint presentation (8–10 minutes) explaining your design process, referencing sources, and showing visuals clearly. Ensure your camera and screen are on while presenting.

- When quoting or referencing, show the highlighted text from the source to demonstrate authenticity.

R&U5: Identify the key elements needed for a simple XR experience. R&U6: Describe common methods that keep users engaged in XR.				
	PROFICIENT [8-10 Marks]	NEARING PROFICIENT [4-7 Marks]	NOVICE [1-3 Marks]	INADEQUATE [0 MARKS]
[10 Marks]	Clearly identifies and applies key XR engagement design and usability principles from credible sources. Demonstrates strong understanding of user experience, interaction design, and comfort/safety factors. Presents a creative and feasible XR scenario supported by professional diagrams and clear explanations. Video presentation is confident and well-structured.	Identifies relevant design principles and applies them with partial justification. Scenario is clear but lacks depth or strong linkage to guidelines. Visual materials are adequate but may miss consistency or clarity. Presentation is mostly clear.	Mentions general XR ideas with limited or inaccurate reference to usability principles. Scenario and visuals are underdeveloped or inconsistent. Lacks clear understanding of XR experience design. Presentation unclear or incomplete.	Missing or irrelevant work

Task 3

In this task, you are to design and develop an interactive VR farm-building game using Unity's XR Interaction Toolkit & Simulator. The player must be able to move freely, build a farm, plant and grow crops, harvest produce, and sell it for coins. Below is a list of functionalities that need to be implemented. You must also produce a screen-recorded video (6–8 minutes) explaining a gameplay demonstration of all required functionalities one by one. You must also explain the code conducted for every functionality. Be clear in your description. If a functionality is not implemented, then clearly state that it was not implemented. You are also required to submit the Unity project (without the library folder). Game should be similar to: https://www.youtube.com/watch?v=l8N5-ooWMs4

For the video recording make use of this PowerPoint presentation: XR_Section2_Task3_Template.pptx

A&A2: Implement spatial input so a user can grab, press or move a virtual item. A&A5: Experiment with variations of game elements to create distinctive gameplay experiences. E&C3: Develop a coherent game-code structure that supports the incorporated mechanics and engagement goals.		
Functionality ID	**Description**	**Max Mark**
G1	VR setup uses hand skins for interaction instead of default controller skins.	1
G2	The player can move and teleport smoothly within the environment using standard VR locomotion controls.	1
G3	The fence placement system allows the player to rotate fences in 90° increments using the joystick or input before placing them in position.	2
G4	The player can grab and place objects (e.g., seeds, crates, tools) in the correct predefined places using sockets.	2
G5	Tillage plots accept seeds or plantable items; once placed, these trigger a growth process that progresses automatically over time.	2
G6	Plants visually grow through multiple stages with smooth transitions or model swaps.	2
G7	Harvesting mechanic implemented. Fruit detaches and can be collected when the player is nearby using grab.	4
G8	The inventory system tracks harvested items (type and count) in real time.	2
G9	Shop or selling menu allows the player to sell harvested items; coins or points increase based on the quantity sold, displayed in a user interface.	2
G10	The player can interact with farm animated animals.	2
G11	The player can drive a farm vehicle to move around. The user can rotate the vehicle steering.	4

Task 4

In this task, you are to develop a new Unity VR project using the XR Interaction Toolkit (XR Interaction Simulator prefab) to record and replay user behaviour. The goal is to create a small research-ready prototype that captures user actions (movement, controller input, teleportation, and grabbing) and can replay them accurately for later analysis. You can make use of grey boxing assets. Submit the Unity project without the library folder.

You are required to:

- Integrate and make use of the XR Interaction Simulator prefab provided by Unity.
- Implement a Record button and a Replay button with the following functionality:
    1. Record: when pressed, begins logging user data to a file (e.g., JSON, CSV, or binary).
    2. Replay: loads the saved data and reproduces the recorded session in the scene (e.g., moving around, rotating controllers, teleporting and grabbing objects).
- The recording must include:
    - Movement of the XR Origin (head position)
    - Movement and rotation of both controllers

- Teleportation events
- Grabbing and release actions

- During replay, the XR Interaction Simulator buttons must animate correctly (e.g., when the user pressed “G” for grab, the button animation should trigger exactly as in the original session).
- Present your working solution through a screen-recorded video (6–8 minutes) showing:
  - How the system records data
  - How the data file is stored and loaded
  - The replay in action with visible simulator button animations
  - A short explanation of the key scripts and logic used
- It is highly recommended that you make use of the singleton: XRInteractionSimulator.instance;

R&U4: Identify input data that must be captured for a chosen interaction. A&A3: Investigate the user experience while the application is running.				
	PROFICIENT [8-12 Marks]	NEARING PROFICIENT [3-7 Marks]	NOVICE [1-2 Marks]	INADEQUATE [0 MARKS]
[12 Marks]	Successfully implements full record and replay system capturing XR Origin movement, controller rotation, teleportation, and grab actions. Replay reproduces interactions accurately, including simulator button animations. Code is well-structured and is clearly explained in the video. Presentation is professional and demonstrates a strong understanding of XR input simulation.	Records and replays most interactions correctly but with minor syncing or animation issues. Code shows understanding of main concepts with partial implementation of simulator events. Video explains logic with minor clarity gaps. Presentation is present but could be improved.	Attempts to record or replay but lacks functionality or accuracy. Key interactions (teleport, grab) missing or broken. Limited code explanation. Presentation is present but could be improved.	Missing or irrelevant work

--END OF ASSIGNMENT—